

AgriConnect Backend - Task Assignments

Team: Kelvin (Lead) | Afia | Hanz

Stack: Node.js, Express.js, TypeScript, Prisma, AWS

Database: Handled by DB Team (not our responsibility)

1. GENERAL INSTRUCTIONS

- We are building the backend API for AgriConnect.
- All API responses must follow this format:
Success: { "success": true, "data": { ... } }
Error: { "success": false, "error": { "code": "CODE", "message": "..." } }
- Every endpoint must have Zod validation schemas for request bodies.
- Work on a separate branch from 'dev' (e.g., feature/A1-jwt-auth) and PR back into 'dev'.

2. KELVIN'S TASKS

Task K1 - Project Scaffolding (Deadline: Day 1)

Initialize the Node.js + Express + TypeScript codebase.

- Create private GitHub repo 'agriconnect-backend' with main and dev branches.
- Install express, typescript, ts-node, prisma, @prisma/client, cors, dotenv, helmet, morgan, winston, express-rate-limit, zod, jsonwebtoken, bcryptjs, qrcode, axios, and testing packages.
- Create folders: config, middleware, modules, services, utils.
- Add .env.example with required keys.
- Build standard response helpers, error middleware, and health check route (GET /api/health).

Task K2 - CI/CD Pipeline & AWS Deploy (Deadline: Day 2)

Set up GitHub Actions to lint, compilation check, and test on PRs.

- Set up AWS Elastic Beanstalk or EC2 hosting.
- Add deployment workflow to auto-deploy to AWS on merge to main.

Task K3 - Prisma ORM & Database Setup (Deadline: Day 3)

Connect app to the DB team's PostgreSQL instance.

- Put connection string in .env, run prisma init, pull schema from database, and generate Prisma client.
- Test queries locally to verify connection is working.

Task K4 - Price Recommendation Engine (Deadline: Day 8)

Build pricing calculation logic.

- Create route GET /api/pricing/recommend (takes crop, region, freshness).
- Fetch MOFA price reference. Calculate ceiling (MOFA * freshness/100) and soft floor (MOFA * 0.6).
- Return range with Zod validations and Swagger documentation.

AgriConnect Backend - Task Assignments

3. AFIA'S TASKS

Task A1 - JWT Authentication System (Deadline: Day 3)

Build endpoints inside `src/modules/auth/`:

- POST `/api/auth/register`: Registers user (farmer/buyer/driver), hashes password with `bcrypt`, creates verification OTP.
- POST `/api/auth/verify-otp`: Verifies matching OTP and expiry, updates status to pending admin approval.
- POST `/api/auth/login`: Validates user, returns Access Token (15m) + Refresh Token (7d).
- POST `/api/auth/refresh`: Takes refresh token and issues new access token.
- POST `/api/auth/logout`: Deletes refresh token from database.

Task A2 - Auth Middleware & guards (Deadline: Day 4)

Build Express route protection:

- `authenticate` middleware: Validates JWT Bearer token and attaches user payload to `req.user`.
- `authorize(...roles)` middleware: Restricts route access to specified roles (farmer/buyer/driver/admin).
- `validate(schema)` middleware: Express helper running `Zod` validation on `req.body`.

Task A3 - User Profile Endpoints (Deadline: Day 6)

Build profile modules inside `src/modules/user/`:

- GET `/api/users/profile`: Returns authenticated user's profile info.
- PATCH `/api/users/profile`: Updates profile fields. Handles role-specific parameters (Farmer farm region/GPS, Buyer business name/address, Driver truck capacity/region/availability).

Task A4 - Paystack Escrow Setup (Deadline: Day 8)

Create sandbox account on Paystack.

- Build `PaystackService` wrapper implementing `initializeTransaction`, `verifyTransaction`, `createTransferRecipient`, and `initiateTransfer`.
- Create webhook receiver endpoint POST `/api/payments/webhook` with SHA-512 verification.

AgriConnect Backend - Task Assignments

4. HANZ'S TASKS

Task H1 - Produce Listing CRUD (Deadline: Day 5)

Build produce listing routes inside src/modules/listing/:

- POST /api/listings: Creates listing. Generates SHA-256 hash of details + QR code (via qrcode package). Saves both to database.
- GET /api/listings: Returns own active farmer listings.
- GET /api/listings/:id: Returns detailed single listing.
- PATCH /api/listings/:id: Edit price, quantity, or image URL.
- DELETE /api/listings/:id: Marks listing as inactive (soft delete).

Task H2 - Marketplace Browse & Filter (Deadline: Day 6)

Build buyer browsing inside src/modules/marketplace/:

- GET /api/marketplace: Paginated list of active listings. Supports crop type, region, freshness score, and quantity filters. Supports sorting by freshness, price, or date.

Task H3 - Tamper-Proof Audit Trail (Deadline: Day 7)

Implement secure transaction logging:

- Create AuditService.log(eventType, entityId, data, userId). Hashes event data and chains it to the previous audit log's hash (hash chain).
- Build GET /api/audit/:entityId route verifying full event sequence integrity.

Task H4 - Arkesel SMS & Firebase Notifications (Deadline: Day 9)

Connect outbound messaging services:

- Build SmsService for Arkesel (sendOtp, sendDriverNotification).
- Build NotificationService wrapping FCM push notifications.
- Plug SmsService into Afia's registration flow.